

# Chapter 9D

## Movement Disorders, Dementia, and Cognitive Neurology

Medvale Internal Medicine - Neurology for Internal Medicine

This chapter is generated from the uploaded neurology segment that covers Parkinson disease, Parkinson-plus syndromes, Huntington disease, tremor, ataxia, restless legs syndrome, Tourette syndrome, dementia, Alzheimer disease, and dementia with Lewy bodies. It is intentionally content-heavy and designed for clinical reasoning plus Graph-RAG extraction.

### 1. Big picture: movement and cognition are pattern-recognition problems

Movement disorders and dementias are often diagnosed from the tempo, body distribution, associated non-motor features, and medication history rather than from one definitive lab test. Parkinson disease is a clinical diagnosis built around bradykinesia plus tremor, rigidity, or postural instability. Tremor is divided by activation state: rest tremor suggests parkinsonism, action/intention tremor suggests cerebellar disease, and postural/kinetic tremor suggests essential or physiologic tremor. Dementia is chronic cognitive decline that interferes with function; delirium is acute fluctuating attention failure, and depression can mimic cognitive decline.

The most useful approach is to ask four questions: Is the movement too little, too much, poorly coordinated, or irresistible? Is cognition impaired acutely or chronically? Are there reversible contributors such as medications, thyroid disease, B12 deficiency, depression, sleep disorder, or normal pressure hydrocephalus? Are there red flags for atypical disease, such as early falls, vertical gaze palsy, severe autonomic dysfunction, hallucinations before motor symptoms, rapid decline, or focal neurologic deficits?

#### Core movement/cognitive syndromes

| Syndrome                 | Pattern                                                                                                                                              | Clinical anchor                                                                                                |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Parkinson disease        | Hypokinetic movement disorder with bradykinesia, rigidity, rest tremor, and gait/postural changes.                                                   | Clinical diagnosis; carbidopa/levodopa is the most effective symptomatic therapy for motor features.           |
| Parkinson-plus syndromes | Parkinsonism with atypical features such as early falls, poor levodopa response, autonomic failure, cerebellar signs, gaze palsy, or early dementia. | Think PSP, MSA, corticobasal degeneration, dementia with Lewy bodies.                                          |
| Essential tremor         | Action/postural tremor, often bilateral, sometimes head or voice tremor.                                                                             | Improves with alcohol in many patients; no bradykinesia or rigidity.                                           |
| Huntington disease       | Chorea, behavioral change, psychiatric symptoms, dementia; autosomal dominant CAG expansion.                                                         | Family history is common but absence does not fully exclude.                                                   |
| Ataxia                   | Gait instability, dysmetria, nystagmus, impaired coordination.                                                                                       | Localize to cerebellum, vestibular system, sensory pathways, toxins, vitamin deficiency, or inherited disease. |
| Dementia                 | Progressive cognitive decline with preserved consciousness until late.                                                                               | Differentiate Alzheimer, vascular, Lewy body, frontotemporal, and reversible mimics.                           |

### 2. Parkinson disease: clinical diagnosis and illness script

Parkinson disease is the most common hypokinetic movement disorder. The typical onset is after age 50, and diagnosis is based on clinical pattern rather than a routine blood test. The classic motor pattern is TRAP: tremor at rest, rigidity, akinesia/bradykinesia, and postural instability. The source emphasizes pill-rolling rest tremor, cogwheel rigidity, bradykinesia, masked facies, micrographia, shuffling gait, and difficulty initiating steps.

Pathologically, idiopathic Parkinson disease is associated with degeneration of dopaminergic neurons in the substantia nigra pars compacta and Lewy bodies containing alpha-synuclein. Clinically, however, many patients also have non-motor symptoms such as constipation, REM sleep behavior disorder, anosmia, depression, anxiety, autonomic dysfunction, urinary symptoms, sleep fragmentation, fatigue, pain, and cognitive impairment. These non-motor features can precede motor diagnosis and strongly affect quality of life.

## Parkinson disease clinical features

| Domain              | Findings                                                                                              | Clinical use                                                       |
|---------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Tremor              | Resting, pill-rolling, asymmetric; often decreases with voluntary movement.                           | Rest tremor supports parkinsonism but absence does not exclude PD. |
| Bradykinesia        | Slowness, reduced amplitude with repetitive tasks, reduced arm swing, micrographia.                   | Required for most clinical diagnostic frameworks.                  |
| Rigidity            | Lead-pipe or cogwheel resistance through range of motion.                                             | Often asymmetric early.                                            |
| Gait/posture        | Shuffling gait, stooped posture, en bloc turning, freezing, postural instability later.               | Early falls suggest atypical parkinsonism.                         |
| Face/voice/swallow  | Masked facies, hypophonia, dysarthria, dysphagia, drooling.                                           | Predicts aspiration and nutrition concerns in advanced disease.    |
| Autonomic/non-motor | Constipation, orthostasis, urinary dysfunction, erectile dysfunction, sleep disorders, mood symptoms. | May drive morbidity more than tremor.                              |

## Parkinson treatment framework

Treatment is symptomatic and individualized by age, functional impairment, comorbidity, cognitive status, work demands, side-effect risk, and patient goals. AAN guidance on early Parkinson motor symptoms states that levodopa provides superior motor benefit compared with dopamine agonists or MAO-B inhibitors, while dopamine agonists are more associated with impulse-control disorders, sleepiness, hallucinations, and edema. Levodopa remains the most effective medication for motor symptoms, but long-term therapy can lead to motor fluctuations and dyskinesias.

## Parkinson therapies

| Therapy                | Role                                                                                                      | Important adverse effects / cautions                                                                                |
|------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Carbidopa/levodopa     | Most effective symptomatic therapy for bradykinesia and rigidity; also helps tremor in many patients.     | Nausea, orthostasis, hallucinations, dyskinesias, wearing-off, on-off fluctuations.                                 |
| Dopamine agonists      | Can help early disease or adjunct therapy; may delay levodopa in selected younger patients.               | Impulse-control disorders, sleep attacks, hallucinations, edema; caution in older or cognitively impaired patients. |
| MAO-B inhibitors       | Mild symptomatic benefit; adjunct for wearing-off.                                                        | Drug interactions, insomnia, nausea; modest effect.                                                                 |
| Amantadine             | May help dyskinesia and mild symptoms.                                                                    | Confusion, hallucinations, livedo reticularis, edema; renal dosing concerns.                                        |
| Anticholinergics       | Occasionally useful for tremor-predominant disease in younger patients.                                   | Cognitive impairment, urinary retention, constipation, glaucoma; avoid in older adults.                             |
| Deep brain stimulation | Selected patients with medication-responsive PD and disabling tremor, dyskinesias, or motor fluctuations. | Requires careful selection; not for uncontrolled dementia or severe psychiatric disease.                            |

Medication-induced parkinsonism is common and should be checked before labeling a patient with idiopathic Parkinson disease. Dopamine-blocking antipsychotics, metoclopramide, prochlorperazine, and reserpine-like agents can cause or worsen parkinsonian features.

## 3. Parkinson-plus syndromes and red flags

Parkinson-plus syndromes are disorders with parkinsonism plus additional findings that do not fit typical idiopathic Parkinson disease. The uploaded source highlights progressive supranuclear palsy, multiple system atrophy, and corticobasal degeneration. These syndromes often have poor levodopa response and early disability. Red flags include early falls, vertical gaze palsy, severe autonomic failure, cerebellar signs, pyramidal signs, early dementia, prominent hallucinations before dopaminergic therapy, and marked asymmetry with apraxia or cortical sensory loss.

## Atypical parkinsonism

| Syndrome                       | Clues                                                                                                                      | What separates it from typical PD                              |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Progressive supranuclear palsy | Early falls, axial rigidity, dysphagia, cognitive change, vertical gaze palsy.                                             | Look for impaired downgaze and early postural instability.     |
| Multiple system atrophy        | Parkinsonism plus autonomic failure, urinary dysfunction, erectile dysfunction, orthostasis; may have cerebellar features. | Severe early autonomic dysfunction and poor levodopa response. |

| Syndrome                  | Clues                                                                                         | What separates it from typical PD                                |
|---------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Corticobasal degeneration | Marked asymmetric rigidity/dystonia, apraxia, cortical sensory loss, alien limb, myoclonus.   | Cortical signs with asymmetric movement disorder.                |
| Dementia with Lewy bodies | Dementia plus visual hallucinations, fluctuations, parkinsonism, REM sleep behavior disorder. | Cognitive/hallucination features early; neuroleptic sensitivity. |
| Drug-induced parkinsonism | Symmetric parkinsonism after dopamine-blocking drug exposure.                                 | Medication timing; may improve after withdrawal.                 |

## 4. Tremor: classify by activation state

Tremor is rhythmic oscillation of a body part. The source table divides common tremors into parkinsonian, cerebellar, and essential patterns. The most useful first question is: does it occur at rest, with posture, with action, or near the target? Rest tremor suggests parkinsonism. Intention tremor that worsens as the finger approaches the target suggests cerebellar disease. Postural/kinetic tremor suggests essential tremor, physiologic tremor, medication effect, hyperthyroidism, caffeine, or withdrawal.

### Tremor differentiation

| Tremor type          | When it appears                                       | Associated clues                                                                      | Treatment direction                                   |
|----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------|
| Parkinsonian         | At rest; often decreases with action.                 | Bradykinesia, rigidity, shuffling gait, masked facies.                                | Treat Parkinson syndrome; levodopa often helps.       |
| Essential            | Posture and action; writing, utensils, arms extended. | Often familial; head/voice tremor; improves with alcohol in many patients.            | Propranolol or primidone when functionally impairing. |
| Cerebellar           | Intention/action; worsens near target.                | Ataxia, dysmetria, nystagmus, scanning speech.                                        | Treat underlying cerebellar lesion/toxin/deficiency.  |
| Enhanced physiologic | Postural, fine, variable.                             | Caffeine, anxiety, hyperthyroidism, beta-agonists, withdrawal.                        | Remove trigger; beta-blocker for selected cases.      |
| Medication/toxin     | Variable rest/action/postural.                        | Lithium, valproate, stimulants, bronchodilators, thyroid hormone, alcohol withdrawal. | Medication review and adjustment.                     |

## 5. Huntington disease and chorea

Huntington disease is an autosomal dominant neurodegenerative disorder caused by CAG repeat expansion in the HTT gene. It classically presents between ages 30 and 50 with chorea, behavioral and psychiatric changes, and progressive cognitive decline. Chorea involves irregular, unpredictable, flowing movements of the face, limbs, trunk, and tongue. The source emphasizes loss of GABA-producing neurons in the striatum, impaired mentation, personality changes, depression, psychosis, and progressive disability.

### Chorea differential

| Cause                 | Clues                                                                      | Testing/management                                                   |
|-----------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------|
| Huntington disease    | Family history, midlife onset, chorea plus psychiatric/cognitive decline.  | Genetic testing with counseling.                                     |
| Medication-induced    | Levodopa, antipsychotic/tardive syndromes, stimulants.                     | Medication history; withdraw/adjust offending drug if possible.      |
| Metabolic/endocrine   | Hyperthyroidism, hyperglycemia, pregnancy-related chorea.                  | Targeted labs.                                                       |
| Autoimmune/infectious | Sydenham chorea, SLE/APS.                                                  | History, exam, inflammatory/autoimmune testing when indicated.       |
| Wilson disease        | Young patient with movement disorder, liver disease, psychiatric symptoms. | Ceruloplasmin, copper studies, slit lamp for Kayser-Fleischer rings. |

## 6. Ataxia: cerebellar, sensory, vestibular, and toxic patterns

Ataxia means impaired coordination and balance. The source divides acquired causes such as alcohol intoxication, vitamin B12 or thiamine deficiency, cerebellar infarction or tumor, demyelinating disease, AIDS, and tertiary syphilis from inherited causes such

as Friedreich ataxia and ataxia-telangiectasia. Bedside localization is central: cerebellar ataxia causes dysmetria and wide-based gait; sensory ataxia worsens with eyes closed because proprioceptive input is lost; vestibular disease causes vertigo and imbalance; frontal gait disorders may mimic ataxia but are driven by initiation and postural control problems.

## Ataxia localization

| Pattern         | Findings                                                                                      | Common causes                                                          |
|-----------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Cerebellar      | Dysmetria, dysdiadochokinesia, intention tremor, nystagmus, scanning speech, wide-based gait. | Stroke, alcohol, tumor, MS, drugs, paraneoplastic disease.             |
| Sensory         | Loss of vibration/proprioception, positive Romberg, stomping gait.                            | B12 deficiency, tabes dorsalis, neuropathy.                            |
| Vestibular      | Vertigo, nausea, nystagmus, veering/falling.                                                  | Vestibular neuritis, Meniere disease, stroke.                          |
| Toxic/metabolic | Global incoordination, sedation, confusion.                                                   | Alcohol, sedatives, anticonvulsants, lithium, Wernicke encephalopathy. |
| Inherited       | Longstanding progression, family history, childhood/young adult onset.                        | Friedreich ataxia, ataxia-telangiectasia.                              |

## 7. Restless legs syndrome and Tourette syndrome

Restless legs syndrome is an urge to move the legs, usually with uncomfortable sensations, worse at rest and in the evening, relieved by movement. It can be associated with iron deficiency, pregnancy, renal failure, neuropathy, and medications. The first step is to check for reversible contributors, especially low ferritin or iron deficiency, before escalating pharmacologic therapy.

Tourette syndrome involves both motor and phonic tics before age 18, persisting for more than one year. Tics are sudden, recurrent, nonrhythmic movements or vocalizations. The source notes association with obsessive-compulsive disorder and that not all patients with tics have Tourette syndrome; seizures, tardive dyskinesia, Huntington disease, and stereotypies should be considered when the pattern is atypical.

### RLS and tic disorders

| Syndrome               | Diagnosis clues                                                                    | Treatment frame                                                                                                        |
|------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Restless legs syndrome | Urge to move legs, worse at rest/evening, relief with movement; sleep disruption.  | Check ferritin/iron; remove triggers; dopamine agonists, alpha-2-delta ligands, or other therapy depending on patient. |
| Tourette syndrome      | Multiple motor tics plus at least one phonic tic, onset before 18, chronic course. | Education; behavioral therapy; clonidine/guanfacine or antipsychotics for impairing tics.                              |
| Tardive dyskinesia     | Choreiform oral-buccal-lingual or limb movements after dopamine-blocking agents.   | Medication review; VMAT2 inhibitor consideration.                                                                      |

## 8. Dementia: syndrome, not one disease

Dementia is progressive decline in cognition that interferes with independence. Consciousness is usually preserved until late stages, which helps separate dementia from delirium. The uploaded source emphasizes increasing age as the most important risk factor and lists Alzheimer disease, vascular dementia, mass lesions, normal pressure hydrocephalus, frontotemporal dementia, dementia with Lewy bodies, neurologic diseases, infections, metabolic disorders, drugs/toxins, and pseudodementia as major considerations.

### Dementia differential

| Type                      | Typical pattern                                                                                                   | Distinguishing clues                                                                        |
|---------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Alzheimer disease         | Insidious memory-predominant decline progressing to language, visuospatial, executive, and functional impairment. | Age, family history, cortical atrophy; diagnosis remains clinical with exclusion of mimics. |
| Vascular dementia         | Stepwise or fluctuating decline after strokes or small-vessel disease.                                            | Focal deficits, gait changes, vascular risk factors, imaging burden.                        |
| Dementia with Lewy bodies | Fluctuating cognition, visual hallucinations, parkinsonism, REM sleep behavior disorder.                          | Neuroleptic sensitivity; hallucinations early.                                              |
| Frontotemporal dementia   | Behavior/personality or language changes earlier than memory loss.                                                | Disinhibition, apathy, compulsions, dietary change, primary progressive aphasia.            |

| Type                          | Typical pattern                                                              | Distinguishing clues                                                                     |
|-------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Normal pressure hydrocephalus | Gait impairment, cognitive decline, urinary incontinence.                    | Ventriculomegaly; gait often first and most prominent.                                   |
| Depression/pseudodementia     | Poor concentration, low mood, sleep/appetite change, 'I don't know' answers. | Potentially reversible; screen and treat mood disorder.                                  |
| Reversible/metabolic          | Variable cognitive slowing or decline.                                       | B12, thyroid, medications, alcohol, sleep apnea, infection, syphilis/HIV when indicated. |

## Dementia workup: practical internal medicine approach

- Establish baseline and time course using collateral history from family/caregivers.
- Screen for delirium first if the change is acute or fluctuating.
- Clarify functional impairment: finances, medications, driving, cooking, appointments, safety, wandering.
- Review medications for anticholinergics, sedatives, opioids, antihistamines, dopamine blockers, anticonvulsants, and polypharmacy.
- Assess mood, sleep, alcohol/substance use, hearing, vision, gait, falls, urinary symptoms, and neurologic focality.
- Initial labs often include CBC, CMP, TSH, B12, and additional tests such as HIV, syphilis, folate, thiamine, or inflammatory markers when history supports.
- Brain imaging is useful when onset is young, rapid, focal, atypical, associated with gait/incontinence, anticoagulation, cancer, trauma, or concern for mass/subdural/NPH.

## 9. Alzheimer disease

Alzheimer disease is the most common cause of dementia and typically begins with episodic memory impairment, difficulty learning new information, and gradual progression over years. Later features include impaired judgment, language disturbance, visuospatial dysfunction, apraxia, delusions, hallucinations, and loss of activities of daily living. The source highlights amyloid plaques, neurofibrillary tangles, neuronal loss, and cerebral atrophy.

Treatment is symptomatic and supportive. Cholinesterase inhibitors such as donepezil, rivastigmine, and galantamine may offer modest symptomatic benefit in Alzheimer disease. Memantine is often used for moderate to severe disease. Management also includes safety planning, caregiver support, exercise, sleep optimization, hearing/vision correction, treatment of mood symptoms, vascular risk control, and advance care planning.

### Alzheimer disease management buckets

| Bucket                 | Examples                                                                            | Goal                                                  |
|------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------|
| Cognitive symptoms     | Cholinesterase inhibitors; memantine in selected moderate-severe cases.             | Modest symptomatic support, not cure.                 |
| Function and safety    | Medication supervision, driving assessment, fall prevention, wandering plan.        | Reduce harm and caregiver crisis.                     |
| Behavioral symptoms    | Pain, sleep, infection, constipation, sensory impairment, environment before drugs. | Treat triggers; avoid unnecessary sedatives.          |
| Comorbidities          | HTN, diabetes, depression, sleep apnea, hearing/vision loss.                        | Preserve cognition/function and reduce delirium risk. |
| Caregiver and planning | Education, support, respite, legal/financial/advance directives.                    | Sustain care and respect patient goals.               |

## 10. Dementia with Lewy bodies

Dementia with Lewy bodies has overlapping features of Alzheimer disease and Parkinson disease, but cognitive fluctuation and hallucinations are especially prominent. The uploaded source notes early visual hallucinations, extrapyramidal features, fluctuating mental status, and sensitivity to neuroleptic agents. Clinically, the core pattern is dementia plus visual hallucinations, fluctuations, parkinsonism, and REM sleep behavior disorder.

Lewy body dementia trap: dopamine-blocking antipsychotics can markedly worsen parkinsonism, sedation, confusion, rigidity, and even precipitate severe neuroleptic sensitivity. Use nonpharmacologic strategies first and choose medications cautiously when psychosis is dangerous.

## Alzheimer vs Lewy body vs Parkinson disease dementia

| Feature                | Alzheimer                                    | Dementia with Lewy bodies                                                       | Parkinson disease dementia                             |
|------------------------|----------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------|
| Initial syndrome       | Memory-predominant cognitive decline.        | Fluctuations, visual hallucinations, attention/executive/visuospatial deficits. | Established Parkinson motor disease precedes dementia. |
| Motor symptoms         | Late or mild.                                | Parkinsonism often early.                                                       | Parkinsonism first, cognition later.                   |
| Hallucinations         | Later.                                       | Early and recurrent, often visual.                                              | Can occur from disease or dopaminergic therapy.        |
| Medication sensitivity | Anticholinergics/sedatives worsen cognition. | High neuroleptic sensitivity.                                                   | Dopamine blockers worsen motor symptoms.               |

## 11. Integrated cases

### Case 1: asymmetric rest tremor

A 66-year-old has a right hand pill-rolling tremor at rest, small handwriting, reduced right arm swing, and cogwheel rigidity.

Reasoning: Idiopathic Parkinson disease is likely. Confirm bradykinesia and functional impairment, review medications for dopamine blockers, and consider carbidopa/levodopa if symptoms affect function.

### Case 2: early falls and vertical gaze palsy

A 71-year-old with parkinsonism falls backward frequently within the first year and has difficulty looking down.

Reasoning: Progressive supranuclear palsy is suggested by early falls and vertical gaze palsy; levodopa response is often limited.

### Case 3: action tremor

A 48-year-old has bilateral hand tremor while writing and holding a cup. No rigidity or bradykinesia. Tremor improves with alcohol.

Reasoning: Essential tremor is likely. Consider propranolol or primidone if functionally impairing.

### Case 4: chorea and psychiatric change

A 42-year-old develops irregular flowing movements, depression, irritability, and worsening work performance. His father died of a similar illness.

Reasoning: Huntington disease is likely; genetic counseling and testing are central.

### Case 5: hallucinations and parkinsonism

A 78-year-old has fluctuating attention, vivid visual hallucinations, and new parkinsonism. Haloperidol causes marked worsening.

Reasoning: Dementia with Lewy bodies is likely; avoid typical antipsychotics when possible.

### Case 6: gait plus cognition plus urinary symptoms

A 74-year-old has magnetic gait, cognitive slowing, and urinary urgency.

Reasoning: Normal pressure hydrocephalus should be considered; brain imaging for ventriculomegaly and specialty evaluation are appropriate.

## 12. Graph-RAG extraction map

### Concept nodes and relationships

| Node               | Class             | Edges to encode                                                                                                                                     |
|--------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Parkinson disease  | Movement disorder | presents_with bradykinesia/rest tremor/rigidity; associated_with Lewy bodies; treated_by carbidopa-levodopa/dopamine agonists/MAO-B inhibitors/DBS. |
| Carbidopa-levodopa | Therapy           | treats Parkinson motor symptoms; adverse_effect dyskinesia/wearing-off/hallucinations/orthostasis.                                                  |

## 12. Graph-RAG extraction map

### Concept nodes and relationships

| Node                          | Class                                 | Edges to encode                                                                                                                                                    |
|-------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parkinson disease             | Movement disorder                     | presents_with bradykinesia/rest tremor/rigidity; associated_with Lewy bodies; treated_by carbidopa-levodopa/dopamine agonists/MAO-B inhibitors/DBS.                |
| Carbidopa-levodopa            | Therapy                               | treats Parkinson motor symptoms; adverse_effect dyskinesia/wearing-off/hallucinations/orthostasis.                                                                 |
| Parkinson-plus syndrome       | Movement disorder category            | includes PSP/MSA/corticobasal degeneration/DLB; suggested_by early falls/autonomic failure/gaze palsy/poor levodopa response.                                      |
| Essential tremor              | Movement disorder                     | presents_with action/postural tremor; improves_with alcohol; treated_by propranolol/primidone.                                                                     |
| Huntington disease            | Neurodegenerative disease             | caused_by CAG repeat expansion; presents_with chorea/psychiatric symptoms/dementia; inherited_as autosomal dominant.                                               |
| Ataxia                        | Neurologic syndrome                   | localizes_to cerebellum/sensory/vestibular/toxic; caused_by stroke/MS/alcohol/B12/thiamine/inherited disorders.                                                    |
| Dementia                      | Cognitive syndrome                    | distinguished_from delirium by chronic progressive course; caused_by Alzheimer/vascular/DLB/FTD/NPH/metabolic/infectious/drug causes.                              |
| Alzheimer disease             | Dementia subtype                      | presents_with progressive memory decline; associated_with amyloid plaques/neurofibrillary tangles; treated_by cholinesterase inhibitors/memantine/supportive care. |
| Dementia with Lewy bodies     | Dementia subtype                      | presents_with fluctuations/visual hallucinations/parkinsonism; has_risk neuroleptic sensitivity.                                                                   |
| Normal pressure hydrocephalus | Potentially reversible dementia mimic | presents_with gait impairment/cognitive decline/urinary incontinence; evaluated_by brain imaging/specialty testing.                                                |

## 13. High-yield traps

- Do not diagnose Parkinson disease without bradykinesia; isolated tremor can be essential, physiologic, medication-induced, thyroid-related, or cerebellar.
- Early falls, vertical gaze palsy, severe autonomic failure, or poor levodopa response should trigger concern for atypical parkinsonism.
- Dopamine agonists can cause impulse-control disorders and hallucinations; they are not benign substitutes for levodopa.
- Anticholinergics may help tremor but can worsen cognition, constipation, urinary retention, and glaucoma risk.
- Huntington disease can present with psychiatric and cognitive symptoms before chorea becomes obvious.
- Treat reversible dementia contributors: medications, depression, sleep disorders, B12 deficiency, thyroid disease, NPH, infection, alcohol, and sensory impairment.
- Do not give typical antipsychotics casually in dementia with Lewy bodies; neuroleptic sensitivity can be severe.
- Ataxia is a localization sign. Decide whether it is cerebellar, sensory, vestibular, frontal, or toxic/metabolic.

## 14. Practice questions

Q1. A patient has rest tremor, bradykinesia, rigidity, and shuffling gait. Most likely syndrome?

Answer: Parkinson disease.

Q2. Which Parkinson medication is most effective for motor symptoms?

Answer: Carbidopa/levodopa.

Q3. A patient with parkinsonism has early falls and vertical gaze palsy. Think of what?

Answer: Progressive supranuclear palsy.

Q4. A bilateral hand tremor worsens with writing and improves with alcohol. Diagnosis?

Answer: Essential tremor.

Q5. Chorea, psychiatric symptoms, dementia, and autosomal dominant family history suggest what?

Answer: Huntington disease.

Q6. Progressive cognitive decline with preserved consciousness until late is called what syndrome?

Answer: Dementia.

Q7. Visual hallucinations, fluctuating cognition, parkinsonism, and neuroleptic sensitivity suggest what?

Answer: Dementia with Lewy bodies.

Q8. Gait disturbance, urinary symptoms, and cognitive decline suggest which potentially reversible dementia mimic?

Answer: Normal pressure hydrocephalus.

## **15. References and clinical cross-check anchors**

This chapter was generated as original educational content from the uploaded movement disorder and dementia source section. Clinical anchors were cross-checked against AAN guidance for dopaminergic therapy in early Parkinson disease, Movement Disorder Society resources on Parkinsonism, and AAN mild cognitive impairment guidance. Local neurology consultation should guide complex diagnostic confirmation, advanced Parkinson therapies, genetic testing, and dementia medication decisions.